

DeCovarT: A Probabilistic and Multidimensional Framework for Cellular Deconvolution in Heterogeneous Biological Samples

Bastien CHASSAGNOL^{1,2}, Gréogry Nuel^{2,3†}, Etienne Becht^{1,2†}

^{1*}Institut De Recherches Internationales Servier (IRIS), FRANCE,
Organization, Street, City, 100190, State, Country.

²LPSM (Laboratoire de Probabilités, Statistiques et Modélisation),
Sorbonne Université, Organization, Street, City, 10587, State, Country.

³Department, Organization, Street, City, 610101, State, Country.

Contributing authors: bastien_chassagnol@laposte.net;
iiauthor@gmail.com; iiiauthor@gmail.com;

[†]These authors contributed equally to this work.

Abstract

Although bulk transcriptomic analyses have greatly contributed to a better understanding of complex diseases, their sensibility is hampered by the highly heterogeneous cellular compositions of biological samples. To address this limitation, computational deconvolution methods have been designed to automatically estimate the frequencies of the cellular components that make up tissues, typically using reference samples of physically purified populations. However, they perform badly at differentiating closely related cell populations.

We hypothesised that the integration of the covariance matrices of the reference samples could improve the performance of deconvolution algorithms. We therefore developed a new tool, DeCovarT, that integrates the structure of individual cellular transcriptomic network to reconstruct the bulk profile. Specifically, we inferred the ratios of the mixture components by a standard maximum likelihood estimation (MLE) method, using the Levenberg-Marquardt algorithm to recover the maximum from the parametric convolutional distribution of our model. We then consider a reparametrisation of the log-likelihood to explicitly incorporate the simplex constraint on the ratios. Preliminary numerical simulations suggest that this new algorithm outperforms previously published methods, particularly when individual cellular transcriptomic profiles strongly overlap.

Purpose: The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. The abstract must not include subheadings (unless expressly permitted in the journal's Instructions to Authors), equations or citations. As a guide the abstract should not exceed 200 words. Most journals do not set a hard limit however authors are advised to check the author instructions for the journal they are submitting to.

Methods: The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. The abstract must not include subheadings (unless expressly permitted in the journal's Instructions to Authors), equations or citations. As a guide the abstract should not exceed 200 words. Most journals do not set a hard limit however, authors are advised to check the author instructions for the journal they are submitting to.

Results: The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. The abstract must not include subheadings (unless expressly permitted in the journal's Instructions to Authors), equations or citations. As a guide the abstract should not exceed 200 words. Most journals do not set a hard limit however authors are advised to check the author instructions for the journal they are submitting to.

Conclusion: The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. The abstract must not include subheadings (unless expressly permitted in the journal's Instructions to Authors), equations or citations. As a guide the abstract should not exceed 200 words. Most journals do not set a hard limit however authors are advised to check the author instructions for the journal they are submitting to.

Keywords: deconvolution, bulk RNASeq, Gaussian, generative model

1 Definitions

- The spectrum is the set of eigenvalues of a matrix.
- The Diag operator keeps only the diagonal elements of a matrix, setting off-diagonal terms as null.
- the condition number of a matrix $\mathbf{X}$ is given by the ratio of the largest eigenvalue over the smallest eigenvalue $\kappa(\mathbf{X}) = \frac{\lambda_{\max}}{\lambda_{\min}}$. A large condition number is a cue of strong multicollinearity (in other words, strong interactions between variables), leading to more complex inversion of matrices and loss of numerical precision.

2 Introduction

The analysis of the bulk transcriptome provided new insights on the mechanisms underlying disease development. However, such methods ignore the intrinsic cellular heterogeneity of complex biological samples, by averaging measurements over several

distinct cell populations. Failure to account for changes of the cell composition is likely to result in a loss of *specificity* (genes mistakenly identified as differentially expressed, while they only reflect an increase in the cell population naturally producing them) and *sensibility* (genes expressed by minor cell populations are amenable being masked by highly variable expression from major cell populations).

Accordingly, a range of computational methods have been developed to estimate cellular fractions, but they perform poorly in discriminating cell types displaying high phenotypic proximity. Indeed, most of them assume that purified cell expression profiles are fixed observations, omitting the variability and intrinsically interconnected structure of the transcriptome. For instance, the gold-standard deconvolution algorithm *CIBERSORT* [1] applies nu-support vector regression (ν -SVR) to recover the minimal subset of the most informative genes in the purified signature matrix. However, this machine learning approach assumes that the transcriptomic expressions are independent.

In contrast to these approaches, we hypothesised that integrating the pairwise covariance of the genes into the reference transcriptome profiles could enhance the performance of transcriptomic deconvolution methods. The generative probabilistic model of our algorithm, *DeCovarT* (Deconvolution using the Transcriptomic Covariance), implements this integrated approach.

3 Results

See online methods for details.

3.1 Methods overview

3.2 Numerical Simulation Study

3.2.1 Toy Model with two genes, and two cell populations

To assert numerically the relevance of accounting for the correlation between expressed transcripts, we designed a simple toy example with two genes and two cell proportions. Hence, using the simplex constraint (3), we only have to estimate one free unconstrained parameter, θ_1 , and then uses the mapping function, defined in Appendix A.3 to recover the ratios in their original space.

We simulated the bulk mixture, $\mathbf{y} \in \mathcal{M}_{G \times N}$, for a set of artificial samples $N = 500$, with the following generative model:

- We have tested two levels of cellular ratios, one with equi-balanced proportions ($\mathbf{p} = (p_1, p_2 = 1 - p_1) = (\frac{1}{2}, \frac{1}{2})$) and one with highly unbalanced cell populations: $\mathbf{p} = (0.95, 0.05)$.
- Then, each purified transcriptomic profile is drawn from a multivariate Gaussian distribution. We compared two scenarios, playing on the mean distance of centroids, respectively $\mu_{.1} = (20, 22), \mu_{.2} = (22, 20)$ and $\mu_{.2} = (20, 40), \mu_{.2} = (40, 20)$ and building the covariance matrix, $\Sigma \in \mathcal{M}_{2 \times 2}$ by assuming equal individual variances for each gene (the diagonal terms of the covariance matrix,

General
explanation
of
DecovarT's
optimi-
sation
framework

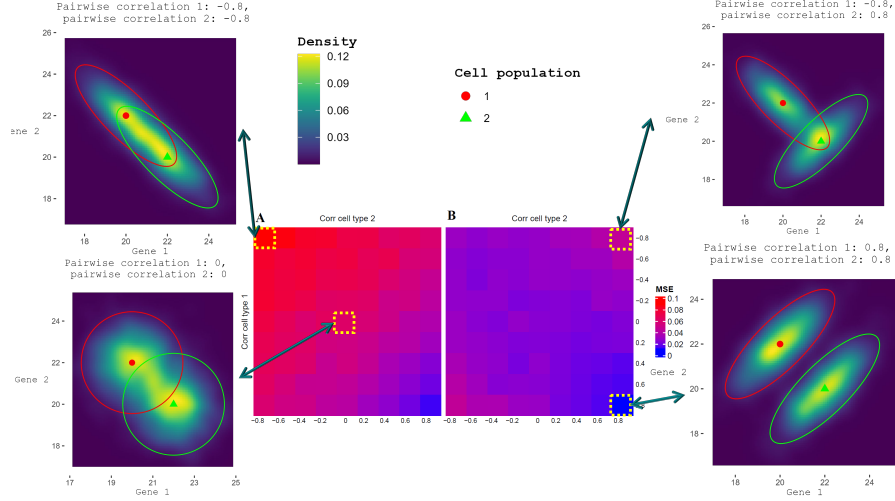


Fig. 1 We used the package [ComplexHeatmap](#) to display the mean square error (MSE) of the estimated cell ratios, comparing the NNLS output, as implemented in the DeconRNASeq algorithm ([2]), in Panel **A**, with our newly implemented DeCovarT algorithm, in Panel **B**. The lower the MSE, the least noisy and biased the estimates. In addition, we added the two-dimensional density plot for the intermediate scenario, for which each population is parameterised by a diagonal covariance matrix, and the most extreme scenarios (those with the highest correlation between genes). The ellipsoids represent for each cell population the 95% confidence region and the red spherical icon and the green triangular icon represent respectively the centroids (average expression of gene 1 and gene 2) of cell population 1 and cell population 2.

$\text{Diag}(\Sigma_1) = \text{Diag}(\Sigma_2) = \mathbf{I}_2$) but varying the pairwise correlation between gene 1 and gene 2, $\text{Cov}[x_{1,2}]$, on the following set of values: $\{-0.8, -0.6, \dots, 0.8\}$ for each of the cell population.

- As stated in 2, we assume that the bulk mixture, $\mathbf{y}_i$ could be directly reconstructed by summing up the individual cellular contributions weighted by their abundance, without additional noise.

We compared the performance of DeCovarT algorithm with DeconRNASeq algorithm ([2]).

Even with a limited toy example including two cell populations characterised only by two genes, we observe that the overlap was a good proxy of the quality of the estimation: the less the overlap between the two cell distributions, the better the quality of the estimation 1.

To evaluate the biological and statistical interest of DeCovarT, we then expand the simulation framework, by encompassing a larger number of cell types, genes, and testing the sensitivity of the model by voluntarily including noise in the benchmark evaluation.

3.2.2 Scaling numerical simulations using methods for generating synthetic networks

The general framework for generating realistic simulations of single-cell profiles while modelling interactions between genes comprises the following steps. First, generate the structure (skeleton of the graph, namely identification of the edges's coverage), represented as a binary adjacency matrix. Then, shift the binary network into a weighted structure, with different interaction weights between genes (as available in a precision matrix), normalised between -1 and 1 -; each cell population exhibits its own distribution profile, parametrised by the simulated network structure. Finally, reflect the heterogeneity of cell composition in biological tissues by drawing cellular proportions from a Dirichlet distribution.

Network generation

Un vaste pan de la littérature exploré la génération de random and synthetic networks, each lying with its own assumptions. The most popular approaches are the Erdős-Rényi model (no particular structure), the preferential attachment model which assumes a scale-free distribution of the degree of the network, and affiliation model, in which local structures prevail (see details in A). However, all these models return a binary adjacency matrix, encoding only for the presence or absence of an interaction.

The advantage of precision matrices precisely lies in the possibility of weighting edges by the strength of interaction, a higher absolute value depicting a stronger regulatory effect. Besides, the precision matrix should have the same sparsity level as the adjacency matrix, while being positive-definite. Inspiring from [3] and [blue](#), we enforce these two constraints with 1:

$$\tilde{\Omega} = \mathbf{A} \times v, \quad \Omega = \tilde{\Omega} + \text{Diag} \left(\left| \min \left(\text{Spectrum}(\tilde{\Omega}) \right) \right| + u \right) \quad (1)$$

with u and v two scalar hyperparameters controlling the degree of connectivity in the graph. Precisely, v controls the magnitude of the interactions, with higher v inducing larger off-diagonal terms, while the second term of the equation $\text{Diag} \left(\min \left(\left| \text{Spectrum}(\tilde{\Omega}) \right| \right) + u \right)$ guarantees the strict positiveness of the matrix (see 3). Accordingly, a fine balance must be found between the two hyperparameters, with a smaller v implying a more diagonal and well-conditioned matrix, while a larger v implies a stronger domination of the off-diagonal, interaction terms, and a larger spreading out of the spectrum.

Of note, we would have considered other simulation framework for generating interesting Gaussian distribution scenarios. For instance, the Toeplitz structure, modelling an exponential decay with distance between variables, is commonly used for generating structured covariance matrices. Indeed, it's controlled by a single parameter ρ , making it highly interpretable, and easy to compute with the Cholesky decomposition. However, since its primary focus is on the covariance, the framework can only be used to encode local dependencies (and not global, accounting for all higher-order interactions, as for the precision matrix) [4, 5].

add visual with eigenvalues plots to demonstrate affine invariance of the spectrum

The **mixSim** package, developed by [6], and the **clusterGeneration**, developed by [7], are other interesting alternatives for generating random clusters of cell populations, each modelled by a multivariate Gaussian distributions. However, **mixSim** plays on the global level of overlap between cell-type specific distributions, while **clusterGeneration** twists the level of degree separation between clusters, as quantified by a variant of the adjusted Rand Index.

3.3 Evaluation of robustness for DeCovarT

- Add Gaussian white noise as a white-error term
- Evaluate sensibility to other types of statistical distributions [8]
- Dropout rate, as in MuSiC [8]
- Incomplete single-cell references, with missing cell populations [8]
- Evaluate tolerance for different single-cell RNASeq protocols [8]

Sensitivity analyses, when the underlying distribution of counts is not Gaussian

4 Methods

4.1 DeCovarT model set-up

In this section, we recall the paradigm underlying most deconvolution algorithms, relating the linear transcriptomic expression in bulk tissues, and cell type-specific gene expressions. To that end, we introduce the following notations:

- $(\mathbf{y} = (y_{gi}) \in \mathbb{R}_+^{G \times N})$ is the total bulk transcriptomic expression, measured in N individuals.
- $\mathbf{X} = (x_{gj}) \in \mathcal{M}_{\mathbb{R}^{G \times J}}$ the signature matrix of the mean expression of G genes in J purified cell populations.
- $\mathbf{p} = (p_{ji}) \in]0, 1[^{J \times N}$ the unknown relative proportions of cell populations in N samples

As in most traditional deconvolution models, we assume that the total bulk expression can be reconstructed by summing the individual contributions of each cell population weighted by its frequency, as stated explicitly in the following linear matricial relationship (2):

$$\mathbf{y} = \mathbf{X} \times \mathbf{p} \quad (2)$$

In addition, we consider unit simplex constraint on the cellular ratios, $\mathbf{p}$ (3):

$$\begin{cases} \sum_{j=1}^J p_j = 1 \\ \forall j \in \tilde{J} \quad p_j \geq 0 \end{cases} \quad (3)$$

4.2 Model assumptions

In contrast to standard linear regression models, we relax in the DeCovarT modelling framework the *exogeneity* assumption, by considering the set of covariates $\mathbf{X}$

as random variables rather than fixed measures, in a process close to the approach of DSection algorithm and DeMixt algorithms. However, to our knowledge, we are the first to weaken the independence assumption between observations by explicitly considering a multivariate distribution and integrating the intrinsic covariance structure of the transcriptome of each purified cell population.

To do so, we conjecture that the G -dimensional vector $\mathbf{x}_j$ characterising the transcriptomic expression of each cell population follows a multivariate Gaussian distribution, given by 4:

$$\text{Det}(2\pi\mathbf{\Sigma}_j)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{x}_j - \boldsymbol{\mu}_{\cdot j})\mathbf{\Sigma}_j^{-1}(\mathbf{x}_j - \boldsymbol{\mu}_{\cdot j})^\top\right) \quad (4)$$

and parametrised by:

- $\boldsymbol{\mu}_{\cdot j}$, the mean purified transcriptomic expression of cell population j
- $\mathbf{\Sigma}_j$, the covariance matrix of each cell population, constrained to be *positive-definite* (see Appendix A.1). Precisely, we retrieve it from inferring its inverse, known as the precision matrix, through the gLasso [9] algorithm. We define $\boldsymbol{\Theta}_j \equiv \mathbf{\Sigma}_j^{-1}$ the corresponding *precision matrix*, whose inputs, after normalisation, store the partial correlation between two genes, conditioned on all the others. Notably, pairwise gene interactions whose corresponding off-diagonal terms in the precision matrix are null are considered statistically spurious, and discarded.

To derive the log-likelihood of our model, first we *plugged-in* the mean and covariance parameters $\zeta_j = (\boldsymbol{\mu}_{\cdot j}, \mathbf{\Sigma}_j)$ estimated for each cell population in the previous step. Then, setting $\boldsymbol{\zeta} = (\boldsymbol{\mu}, \mathbf{\Sigma})$, $\boldsymbol{\mu} = (\boldsymbol{\mu}_{\cdot j})_{j \in \tilde{J}} \in \mathcal{M}_{G \times J}$, $\mathbf{\Sigma} \in \mathcal{M}_{G \times G}$ the known parameters and $\mathbf{p}$ the unknown cellular ratios, we show that the conditional distribution of the observed bulk mixture, conditioned on the individual purified expression profiles and their ratios in the sample, $\mathbf{y} | (\boldsymbol{\zeta}, \mathbf{p})$, is the convolution of pairwise independent multivariate Gaussian distributions. Using the *affine invariance* property of Gaussian distributions, we can show that this convolution is also a multivariate Gaussian distribution, given by 5.

$$\mathbf{y} | (\boldsymbol{\zeta}, \mathbf{p}) \sim \mathcal{N}_G(\boldsymbol{\mu}\mathbf{p}, \mathbf{\Sigma}) \text{ with } \boldsymbol{\mu} = (\boldsymbol{\mu}_{\cdot j})_{j \in \tilde{J}}, \quad \mathbf{p} = (p_1, \dots, p_J) \text{ and } \mathbf{\Sigma} = \sum_{j=1}^J p_j^2 \mathbf{\Sigma}_j \quad (5)$$

The directed graphical models associated with this framework are shown in Fig. 2: panel **a** recalls the classical linear deconvolution template; panel **b** is the unconstrained DeCovarT generative model; panel **c** adds the regularised soft-max mapping $\boldsymbol{\psi}$ that enforces the unit-simplex constraint on $\mathbf{p}$.

In the next section, we provide an explicit formula of the log-likelihood of our probabilistic framework, its gradient and hessian, which in turn can be used to retrieve the MLE of our distribution.

4.3 Derivation of the log-likelihood

From 5, the conditional log-likelihood is readily computed and given by 6:

$$\ell_{\mathbf{y}|\boldsymbol{\zeta}}(\mathbf{p}) = C + \log \left(\text{Det} \left(\sum_{j=1}^J p_j^2 \boldsymbol{\Sigma}_j \right)^{-1} \right) - \frac{1}{2} (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \left(\sum_{j=1}^J p_j^2 \boldsymbol{\Sigma}_j \right)^{-1} (\mathbf{y} - \mathbf{p}\boldsymbol{\mu}) \quad (6)$$

4.4 First and second-order derivation of the unconstrained DeCovarT log-likelihood function

The stationary points of a function and notably maxima, are given by the roots (the values at which the function crosses the x -axis) of its gradient, in our context, the vector: $\nabla \ell : \mathbb{R}^J \rightarrow \mathbb{R}^J$ evaluated at point $\nabla \ell(\mathbf{p}) :]0, 1[^J \rightarrow \mathbb{R}^J$. Since the computation is the same for any cell ratio p_j , we give an explicit formula for only one of them (7):

$$\begin{aligned} \frac{\partial \ell_{\mathbf{y}|\boldsymbol{\zeta}}(\mathbf{p})}{\partial p_j} &= \frac{\partial \log(\text{Det}(\boldsymbol{\Theta}))}{\partial p_j} - \frac{1}{2} \left[\frac{\partial (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top}{\partial p_j} \boldsymbol{\Theta} (\mathbf{y} - \mathbf{p}\boldsymbol{\mu}) + (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \frac{\partial \boldsymbol{\Theta}}{\partial p_j} (\mathbf{y} - \mathbf{p}\boldsymbol{\mu}) + (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \frac{\partial (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})}{\partial p_j} \right] \\ &= -\text{Tr} \left(\boldsymbol{\Theta} \frac{\partial \boldsymbol{\Sigma}}{\partial p_j} \right) - \frac{1}{2} \left[-\boldsymbol{\mu}_{\cdot j}^\top \boldsymbol{\Theta} (\mathbf{y} - \mathbf{p}\boldsymbol{\mu}) - (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \frac{\partial \boldsymbol{\Sigma}}{\partial p_j} \boldsymbol{\Theta} (\mathbf{y} - \mathbf{p}\boldsymbol{\mu}) - (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \boldsymbol{\mu}_{\cdot j} \right] \\ &= -2p_j \text{Tr}(\boldsymbol{\Theta} \boldsymbol{\Sigma}_j) + (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \boldsymbol{\mu}_{\cdot j} + p_j (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \boldsymbol{\Sigma}_j \boldsymbol{\Theta} (\mathbf{y} - \mathbf{p}\boldsymbol{\mu}) \end{aligned} \quad (7)$$

Since the solution to $\nabla(\ell_{\mathbf{y}|\boldsymbol{\zeta}}(\mathbf{p})) = 0$ is not closed, we had to approximate the MLE using iterated numerical optimisation methods. Some of them, such as the Levenberg–Marquardt algorithm, require a second-order approximation of the function, which needs the computation of the Hessian matrix. Deriving once more 7 yields the Hessian matrix, $\mathbf{H} \in \mathcal{M}_{J \times J}$ is given by:

$$\begin{aligned} \mathbf{H}_{i,i} &= \frac{\partial^2 \ell}{\partial p_i^2} = -2 \text{Tr}(\boldsymbol{\Theta} \boldsymbol{\Sigma}_i) + 4p_i^2 \text{Tr}((\boldsymbol{\Theta} \boldsymbol{\Sigma}_i)^2) - 2p_i (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \boldsymbol{\Sigma}_i \boldsymbol{\Theta} \boldsymbol{\mu}_{\cdot i} - \boldsymbol{\mu}_{\cdot i}^\top \boldsymbol{\Theta} \boldsymbol{\mu}_{\cdot i} - \\ &\quad 2p_i (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \boldsymbol{\Sigma}_i \boldsymbol{\Theta} \boldsymbol{\mu}_{\cdot i} - (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} (4p_i^2 \boldsymbol{\Sigma}_i \boldsymbol{\Theta} \boldsymbol{\Sigma}_i - \boldsymbol{\Sigma}_i) \boldsymbol{\Theta} (\mathbf{y} - \mathbf{p}\boldsymbol{\mu}), \quad i \in \tilde{\mathcal{J}} \\ \mathbf{H}_{i,j} &= \frac{\partial^2 \ell}{\partial p_i \partial p_j} = 4p_j p_i \text{Tr}(\boldsymbol{\Theta} \boldsymbol{\Sigma}_j \boldsymbol{\Theta} \boldsymbol{\Sigma}_i) - 2p_i (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \boldsymbol{\Sigma}_i \boldsymbol{\Theta} \boldsymbol{\mu}_{\cdot j} - \boldsymbol{\mu}_{\cdot i}^\top \boldsymbol{\Theta} \boldsymbol{\mu}_{\cdot j} - \\ &\quad 2p_j (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \boldsymbol{\Sigma}_j \boldsymbol{\Theta} \boldsymbol{\mu}_{\cdot i} - 4p_i p_j (\mathbf{y} - \mathbf{p}\boldsymbol{\mu})^\top \boldsymbol{\Theta} \boldsymbol{\Sigma}_i \boldsymbol{\Theta} \boldsymbol{\Sigma}_j \boldsymbol{\Theta} (\mathbf{y} - \mathbf{p}\boldsymbol{\mu}), \quad (i, j) \in \tilde{\mathcal{J}}^2, i \neq j \end{aligned} \quad (8)$$

in which the coloured sections pair one by one with the corresponding coloured sections of the gradient, given in 7.

Matrix calculus can largely ease the derivation of complex algebraic expressions, thus we remind in Appendices A.1 and A.2 relevant matrix properties and derivations. The numerical consistency of these derivatives was asserted with the **numDeriv** package, using the more stable Richardson's extrapolation ([10]).

However, the explicit formulas for the gradient and the hessian matrix of the log-likelihood function, given in 7 and 8 respectively, do not take into account the simplex

constraint assigned to the ratios. While some optimisation methods use heuristic methods to solve this problem, we consider alternatively a reparametrised version of the problem, detailed comprehensively in Appendix A.3.

5 Discussion

While other approaches, such as DeMix and DeMixT, enabled, using a sound statistical framework, to jointly estimate cellular portions and distributions of gene expression per sample and cell population, DeCovarT is the first deconvolution algorithm which explicitly handles the unit-simplex constraint, addressing the intrinsic compositional nature of cellular composition, while integrating interactions between genes.

Through extensive numerical simulations, playing on varying levels of gene

Hence, it provides a strong basis to further derive statistical tests to assert whether the proportion of a given cell population differs significantly between two distinct biological conditions.

5.1 Limitations of the method

5.1.1 Extend the Simulation Framework

The next phase of our evaluation involves real-world experiments, encompassing both blood and solid tumoral samples. To that end, we could start from the Kassandra benchmark, by [11]. This large-scale project evaluates the performance of five established gold-standard and signature-based deconvolution algorithms, including EPIC [12], CIBERSORT [1], CIBERSORTx [13], quanTIseq [14], and ABIS [15]. The evaluation involves deconvolving six publicly available datasets annotated with both flow cytometry and bulk RNA-seq expression.

Enhanced Inference and Integration of Co-Expression Networks

All the popular differential gene expression analyses, such as *limma + voom* ([16]), *EdgeR* ([17]) and *Deseq2* ([18]), tend to overlook gene-gene interactions, comparing independently the expression between two conditions for each gene. The usual univariate approach of DGEAs additionally underlies the need of adjusting p -values, as numerous genes are examined simultaneously, and without accounting for interactions between them, the probability of observing false positives increases.

To account for correlations among observations, two consecutive papers, by [19] and [20], present an innovative Bayesian framework which models proteomic expression across diverse biological conditions as multivariate Gaussian distributions. Insightful discussions with the main author, Marie Chion, suggest a straightforward extension of the method to transcriptomic expression, given the close relationship between two kinds of omics, both depicting counts.

While the methodology was originally designed to delineate differentially expressed genes between two conditions, the method can be readily extended to incorporate a *one-vs-all strategy*. This extension allows for the identification of markers specific to a particular cell population in comparison to all others. Furthermore, the generative

model aligns closely with our deconvolution framework, leveraging the same distributions to describe cellular omic profiles. Alternatively, differential network approaches, such as INDEED, by [21], implement heuristic and dual-optimisation approaches, finding the sweet spot between maximising the mean differences between purified expression profiles and differentiating the neighbourhood network structure.

The gLasso algorithm used to derive the precision matrix associated to each purified cell profile is subjected to *parameter shrinkage*, like any penalty regularisation approach. Notably, in our setting, shrinkage tends to systematically underestimate the non-zero partial correlations of the precision matrix.

To mitigate this issue, one approach is to incorporate the *support* (indicating non-null inputs), derived from the gLasso output, into a conventional Maximum Likelihood Estimation (MLE) framework. The general concept is to utilising the true "zeros" to impose topological constraints on the final Gaussian Graphical Model (GGM). However, it's important to note that unless the undirected Markov network obtained from the gLasso output is a *chordal graph*, there is usually no straightforward mapping between the two topological spaces.

Finally, the inclusion of prior biological knowledge, such as the strength of relationships between transcription factors, retrieved from Protein-Protein Interaction (PPI) networks, can help reduce the exponential space of undirected graphs to explore.

Enhanced Inference and Integration of Co-Expression Networks

All the methods outlined in 5.1.1 yield a subset of genes that distinguish a particular cell population from all others. However, when we combine these gene subsets, we often end up with a non-scalable signature matrix, exhibiting strong multicollinearity due to redundancy among the identified gene markers.

To further refine the final set of genes able to delineating any cell population included in the signature matrix, *AutoGeneS*, by [22], introduces a greedy genetic approach coupled with a dual optimisation approach¹. Precisely, the loss function involves minimising inter-population correlation while simultaneously maximising the distance of the centroids.

We propose instead of this dual optimisation approach the minimisation of the *global overlap* between the concatenated distributions of the cellular profiles. Indeed, this metric not only captures in a single criterion the combined influence of mean inter-cluster distance and differential network structure in delineating purified cellular expression profiles, but supplies a straightforward score easy to interpret. The *overlap* metric precisely measures the shared probability mass and the degree of concurrence in probability densities. In simpler terms, it quantifies the global probability of incorrectly assigning an expression profile to the wrong cell subtype when utilising a maximum a posteriori approach, with the knowledge of each cellular profile's individual parameters.

- General advantages of bulk transcriptomic paired with cellular deconvolution over alternative transcriptomic technologies

¹In standard approaches that rely on linear regression, the condition number serves as the gold-standard metric for assessing the level of precision of the linear model achievable with the design matrix

Joint Estimation of purified Expression Profiles and Cellular Ratios

We detail in Appendix B a Gibbs sampler to approximate the target distribution, here the joint value of the purified profiles and the cellular ratios. In addition, MCMC sampling allows for straightforward incorporation of prior knowledge, and streamlines the derivation of Maximum a Posteriori (MAP) estimates and *credible intervals*.

Precisely, by coupling Gibbs and Metropolis Hasting samplers, we ensure at each iteration that the estimated parameters adhered to the “balance condition”, an essential property guaranteeing the convergence of MCMC chains to a stationary distribution identifiable to the target distribution.

6 Code availability

6.1 Computational environment

All analyses were performed on ... high-performance computing infrastructure using StdEnv/2020. Core statistical analyses and deconvolution bench-marking were conducted in R (version 4.3.1), employing the following key packages: Seurat (v 4.3.0) for single-cell data processing; limma (v3.58.1), edgeR (v4.0.16), and DESeq2 (v1.42.1) for differential expression and discordant-gene filtering; lme4 (v1.1.35.5) and lmerTest (v3.1.3) for mixed-effects modelling; and tidyverse (v2.0.0) for data manipulation. Fifteen deconvolution methods were benchmarked, including EPIC (v1.1.7), FARDEEP (v1.0.1), MuSiC (v1.0.0), MuSiC2 (v0.1.0), BisqueRNA (v1.0.5), BayesPrism (v2.2.2), DWLS, hspe (v0.1), TOAST (v1.16.0), CSC-DRNA (v1.0.3), CSsingle (v0.1.0), DeconRNASeq (v1.44.0), and PREDE (v1.2.1). CIBERSORTx deconvolution was performed via its web interface (<https://cibersortx.stanford.edu/>). Additional methods, AutoGeneS and GEDIT, were implemented in Python (version 3.10) with numpy, pandas, and scipy.

Visualisation was performed using ggplot2 (v3.5.1), patchwork (v1.2.0), and cowplot (v1.1.3). To ensure reproducibility, all random seeds were fixed.

The package used to generate the simulations and infer ratios from virtual or real biological mixtures with the DeCovarT algorithm is implemented on Github [DeCovarT](#).

7 Author contribution

8 Theorem examples from Nature Communication

A fast exponentiation procedure:

Theorem 1 (Theorem subhead) *Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text.*

Algorithm 1 Calculate $y = x^n$

Require: $n \geq 0 \vee x \neq 0$ **Ensure:** $y = x^n$

```
1:  $y \leftarrow 1$ 
2: if  $n < 0$  then
3:    $X \leftarrow 1/x$ 
4:    $N \leftarrow -n$ 
5: else
6:    $X \leftarrow x$ 
7:    $N \leftarrow n$ 
8: end if
9: while  $N \neq 0$  do
10:  if  $N$  is even then
11:     $X \leftarrow X \times X$ 
12:     $N \leftarrow N/2$ 
13:  else [ $N$  is odd]
14:     $y \leftarrow y \times X$ 
15:     $N \leftarrow N - 1$ 
16:  end if
17: end while
```

Proposition 2 *Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text.*

Example 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Remark 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Definition 1 (Definition sub head) Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text.

Proof of Theorem 1 Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. $\square$

Quoted text example. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

Appendix A Synthetic network simulation

The three most popular approaches for generating random, synthetic networks are:

- The Erdős–Rényi model was first described in [23]: edges are generated independently between each pair of nodes with a fixed probability, leading to a homogeneous random graph.
- In the preferential attachment model ([24, 25]), nodes are iteratively added, favouring connection to the most well-connected nodes. This kind of framework is compliant with the scale-free network property, often looked for in biological systems, and resulting in the emergence of hubs. The degree distribution is notably given by a power-law.
- The affiliation model, also known as the stochastic block model (see [26]), assumes a latent indicator variable assigning each node to a community. The generation of edges is then dependent on group membership, with stronger interactions within a given block.

add a drawing of these three approaches

Theorem 3 (Invariance of the spectrum function by affine transformation) *Let X be any square matrix, and consider the affine transformation:*

$$Y = aX + bI$$

with a and b positive scalars, and I the identity matrix. Then, the eigenvectors of Y are the same as X , and the eigenvalues are linearly transformed, shifted by b and scaled by a (larger range if a is above 1).

Appendix B Optimisation and calculus

B.1 Standard linear deconvolution model

However, in real conditions with technical and environmental variability, strict linearity of the deconvolution does not usually hold. Thus, an additional error term is usually considered, and without further assumption on the distribution of this error term, the usual approach to retrieve the best of parameters is by minimising the squared error term between the mixture expressions predicted by the linear model and the actual observed response. This optimisation task is achieved through the ordinary least squares (OLS) approach (B1),

$$\hat{\mathbf{p}}_i^{\text{OLS}} \equiv \arg \min_{\mathbf{p}_i} \|\hat{\mathbf{y}}_i - \mathbf{y}_i\|^2 = \arg \min_{\mathbf{p}_i} \|\mathbf{X}\mathbf{p}_i - \mathbf{y}_i\|^2 = \sum_{g=1}^G \left(y_{gi} - \sum_{j=1}^J x_{gj} p_{ji} \right) \quad (\text{B1})$$

If we additionally assume that the stochastic error term follows a *homoscedastic* zero-centred Gaussian distribution and that the value of the observed covariates (here, the purified expression profiles) is determined (see the standard linear model representation in the left panel of Fig. 2 and the set of equations describing it in B2),

$$y_{gi} = \sum_{j=1}^J x_{gj} p_{ji} + \epsilon_i, \quad y_{gi} \sim \mathcal{N} \left(\sum_{j=1}^J x_{gj} p_{ji}, \sigma_i^2 \right), \quad \epsilon_i \sim \mathcal{N}(0, \sigma_i^2) \quad (\text{B2})$$

then, the MLE is equal to the OLS, which, in this framework, is given explicitly by B3:

$$\hat{\mathbf{p}}_i^{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}_i \quad (\text{B3})$$

and is known under the the Gauss-Markov theorem 5,.

In all what follows, we suppose that the number of cell types composing the biological sample, J is known. In idealized conditions, standard deconvolution analyses assume that each transcript expression is expressed (without noise, no unknown cellular component, ...) as the linear combination of each raw cell population expression contribution, weighted by their corresponding proportions. Accordingly, the relation linking the global transcriptomic expression, $y_{g,i,c(i)}$, with each individual cell contribution is given, for tissue sample i , under phenotypical condition $c(i)$, is given by B4:

$$y_{g,i,c(i)} = \sum_{j=1}^J x_{gj,c(i)} \times p_{ji} \quad (\text{B4})$$

This yields, for each individual i , a system of G linear equations to solve, that can be represented as the following product of matrix, for a given condition $c(i)$ B5:

$$\mathbf{y}_{i,c(i)} = \mathbf{X}_{c(i)} \times \mathbf{p}_i \quad (\text{B5})$$

In a perfect system without noise as modelled in B5, the number of solutions is given by the ‘‘Rouch  -Capelli’’ theorem [27]. Briefly, uniqueness of the solution implies that the number of genes N is equal to the number of cell types J . Secondly, any expression of an individual gene can not be rewritten as a linear combination of the remaining genes (multicollinearity) expressed in the sample (alternatively, this implies that the reference matrix $\mathbf{X}$ is invertible and of full rank J). If the number of genes measured exceeds those of proportions, the system is said to be *overdetermined* [28].

In real conditions, observed data $\mathcal{D}$ may not fit exactly the system of linear equations B5, which is dealt with by most of the deconvolution algorithms by adding a residual noise parameter ϵ_{gi} to B4. To account for noise, some methods focus on

minimising the squared residual between the observed values $\mathbf{y}_i$ and predicted values $\hat{\mathbf{y}}_i$ (*ordinary least squares* method). In the context of linear regression, we denote $\mathbf{p}$ as the unknown *coefficients* of the linear model to estimate, $\mathbf{Y}$ the *predicted or response variable*, and $\mathbf{X}$ as the design matrix associated to the *independent variables*. Formally, the OLS estimate minimises the following quantity B6:

$$\hat{\mathbf{p}}_i^{\text{OLS}} = \arg \min_{\hat{\mathbf{p}}_i} \|\hat{\mathbf{y}}_i - \mathbf{y}_i\|^2 = \arg \min_{\hat{\mathbf{p}}_i} \|\mathbf{X}\hat{\mathbf{p}}_i - \mathbf{y}_i\|^2 = \sum_{g=1}^G \left(y_{gi} - \sum_{j=1}^J x_{gj} \hat{p}_{ji} \right) \quad (\text{B6})$$

Theorem 4 (Normal Equations) *The **Normal equations**, regardless of the distribution of the error term, supply the following OLS estimate, as solution of B6:*

$$\hat{\mathbf{p}}_i^{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}_i \quad (\text{B7})$$

*whose existence implies that an inverse of the design matrix $\mathbf{X}$ does exist.*²

Instead, a generative model can be used to model the observed data $\mathcal{D}$, in which we use a probability density function to model the distribution of the error term. Classically, the parameter is then retrieved using the *maximum likelihood estimation* (MLE) method. It consists in estimating the distribution parameters by maximising the likelihood, or equivalently the log-likelihood of observed data, $\ell(\theta|\mathcal{D}) = \mathbb{P}_\theta(\mathbf{Y}|\mathbf{X})$, with $\theta = (\mathbf{p}, \boldsymbol{\sigma})$ the vector of parameters, respectively the proportions and standard deviations, to estimate.

Theorem 5 (Gauss-Markov theorem) *Under the following assumptions, **weak exogeneity**: the cell type-specific expression profiles are considered fixed and not drawn from probability distributions. Accordingly, we suppose that the expression of each gene within a cell population is constant. Biologically, these hypotheses imply that the cellular expression profiles are uncorrelated: $1 \leq i \leq J, 1 \leq k \leq J, \forall i \neq k, \text{ cov}(x_{gi}, x_{gk}) = 0$. **homoscedasticity**: the variance associated to the measured error is fixed for the set of genes, formally $\text{var}(\epsilon_{gi}) = \sigma_i^2, \forall g \in \tilde{G}$ **null expected value of the error term**: $1 \leq g \leq G, \mathbb{E}(\epsilon_{gi}) = 0$, implying biologically there's no other generation source for the bulk expression data than the set of cell populations referenced in the signature matrix, $\tilde{J}$. Statistically, this also implies that the variance of the measured gene expression does not depend on the predictors. The two previous assumptions are commonly met by using a white Gaussian distribution to model the residual distribution: $\epsilon_{gi} \sim \mathcal{N}(0, \sigma_i)$ **independence of the predicted values**: $1 \leq i \leq G, 1 \leq j \leq G, \forall i \neq j, \text{ cov}(\epsilon_i, \epsilon_j) = 0$, which implies independence between the bulk measures: $\text{cov}(\mathbf{y}_i, \mathbf{y}_j) = 0$ using the weak exogeneity property.³ then the MLE estimate and the OLS estimate share the following properties:*

²This especially implies that no expression of a cell population can be written as the linear combination of the expression of several children cell lines. In addition with the assumption that the cell populations could be partitioned in non non-overlapping categories, it implies that we can't evaluate simultaneously the ratio of parent and child cell lines.

³The last four hypotheses are generally known as the *Gaussian-Markov* hypotheses. The use of a distribution of null average imply that no additional cell population, acting as a *latent variable*, is present in the bulk mixture.

1. the OLS estimate, as defined in B6 and the MLE estimate, $\hat{\mathbf{p}} = \arg \max(\ell(\theta|X))$ are the same.
2. The MLE estimate is unique and the point for which the global maximum of the log-likelihood function is reached.
3. It is the BLUE (best linear unbiased estimator) estimate, i.e. the unbiased estimator (no expected difference between the estimate and the true value) associated to the lowest variance.

Proof of Gauss-Markov theorem To prove part 1 of 5, we assume additionally independence of the bulk expression between samples: $1 \leq j \leq N, 1 \leq k \leq N, \forall j \neq k, \text{cov}(\mathbf{y}_j, \mathbf{y}_k) = 0$, even though [29] tends to deny this IID assumption. Besides, we suppose that the cellular proportions vary from an individual to another (estimation is performed independently on each sample).

Then, we can show that the conditional distribution of each data point y_{gi} to $\mathbf{x}_{g,c(i)}$ follows a Gaussian distribution B8:

$$\mathbb{P}_{\theta_i}(y_{gic(i)}|\mathbf{x}_{g,c(i)}) = \varphi(y_{gic(i)}|\mathbf{x}_{g,c(i)}, \mathbf{p}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{(y_{gic(i)} - \sum_{j=1}^J x_{gjc(i)} \times p_{ji})^2}{2\sigma_i^2}} \quad (\text{B8})$$

using the closeness property of Gaussian distributions under linear transformation and the independence assumption between the response variable and the residual.

Using the several independence assumptions, the global log-likelihood is given by B9:

$$\begin{aligned} \hat{\theta} &= \arg \max \ell(\theta|\mathcal{D}) \\ &= \log(\mathbb{P}(\mathbf{Y}|\theta, \mathbf{X})) \\ &= \log \left(\prod_{c=1}^C \prod_{i=1}^N \prod_{g=1}^G \mathbb{P}_{\sigma_i, \mathbf{p}_i}(y_{gic(i)}|\mathbf{x}_{g,c(i)}) \right) \\ &= K - G \sum_{i=1}^N \log(\sigma_i) - \sum_{c=1}^C \sum_{i=1}^N \sum_{g=1}^G \left(\frac{(y_{gic(i)} - \sum_{j=1}^J x_{gjc(i)} \times p_{ji})^2}{2\sigma_i^2} \right) \mathbf{1}_{c(i)=c} \end{aligned} \quad (\text{B9})$$

with $K = -\frac{G \times N}{2} \log(2\pi)$, a constant. Finding the values for which the derivative of the log-likelihood cancels yield the same estimate for each individual i as returned by the OLS method B6. The MLE estimate provides additionally an estimate of the standard deviations, as the expected value of sum-of-squared residuals: $\sigma_i^2 = \frac{1}{G} \sum_{g=1}^G \epsilon_{gi}^2$, with $\epsilon_{gi} = y_{gic(i)} - \sum_{j=1}^J x_{gjc(i)} \times p_{ji}$ the estimated residual for each individual.

For part 2, to prove that the estimate $\mathbf{p}$ is indeed the local minima of the OLS problem in B6 or the local maxima of the log-likelihood function B9, we need to differentiate once more to obtain the Hessian matrix and show that it is positive definite, a result proven by the *Gauss-Markov theorem*. $\square$

B.2 Multivariate distributions and basic algebra properties

Definition 2 (Multivariate Gaussian distributions) If random vector $\mathbf{X}$ of size G follows a random multivariate Gaussian distribution, $\mathbf{X} \sim \mathcal{N}_G(\mu, \Sigma)$, then its distribution is given by:

$$\text{Det}(2\pi\Sigma)^{-\frac{1}{2}} \exp \left(-\frac{1}{2}(\mathbf{x} - \mu)\Sigma^{-1}(\mathbf{x} - \mu)^\top \right)$$

in which:

- $\mu = \mathbf{X}$ is the G -dimensional mean vector
- Σ is a $G \times G$ positive-definite [3](#) covariance matrix, whose diagonal terms, $\text{Diag}(\Sigma) = [(\text{Var}[X_{i,j}]), \forall (i,j) \in \tilde{G}^2, i = j]^\top$ are the individual variances of each purified gene transcript in population j and off-diagonal terms, $\Sigma_{i,j} = \text{Cov}[X_i, X_j], \forall (i,j) \in \tilde{G}^2, i \neq j$ are the covariance between variables. We note $\Theta = \Sigma^{-1}$, the inverse of the covariance matrix, called the *precision matrix*.

Remark 2 (Affine invariance property of multivariate GMMs) The two following properties hold for a multivariate Gaussian distribution:

- if $\mathbf{X} \sim \mathcal{N}_G(\mu, \Sigma)$, then $p\mathbf{X}$, with p a constant, follows itself a multivariate Gaussian distribution, given by: $p\mathbf{X} \sim \mathcal{N}_G(p\mu, p^2\Sigma)$
- given two independent random vectors $\mathbf{X}_1 \sim \mathcal{N}_G(\mu_1, \Sigma_1)$ and $\mathbf{X}_2 \sim \mathcal{N}_G(\mu_2, \Sigma_2)$ following a multivariate Gaussian distribution, then the random variable $\mathbf{X}_1 + \mathbf{X}_2$ follows itself the multivariate Gaussian distribution:

$$\mathbf{X} + \mathbf{Y} \sim \mathcal{N}_G(\mu_1 + \mu_2, \Sigma_1 + \Sigma_2)$$

By induction, this property generalises to the sum of J independent random vectors of same dimension $\mathbb{R}^G$.

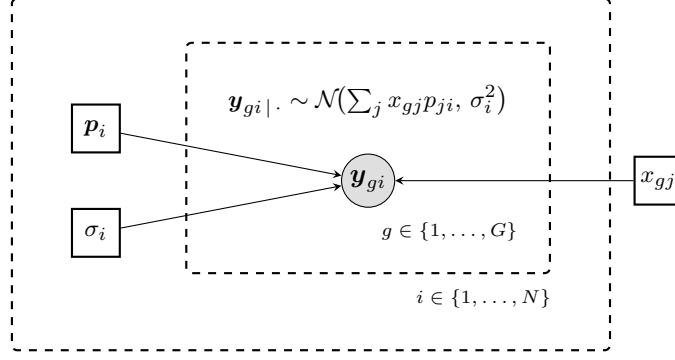
Deriving the characteristic function of the multivariate GMM yields directly results reported in [2](#).

Definition 3 (Definite matrix) A symmetric real matrix $\mathbf{A}$ of rank G is *positive-definite* if:

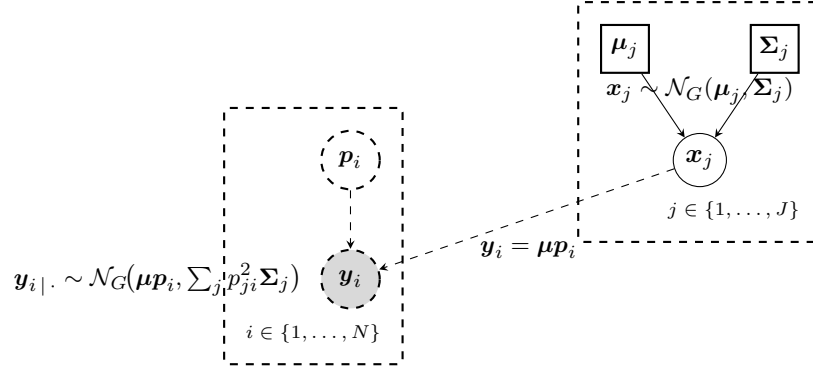
$$\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0, \quad \mathbf{x} \in \mathbb{R}^G \quad (\text{B10})$$

To gain a clearer grasp of the positive-definite constraint imposed on the covariance parameter of a multivariate Gaussian distribution, let's delve into the most straightforward scenario, in which we assume that any of the individual features exhibit pairwise independence. This particular setup is parametrised by a covariance matrix containing exclusively diagonal elements.

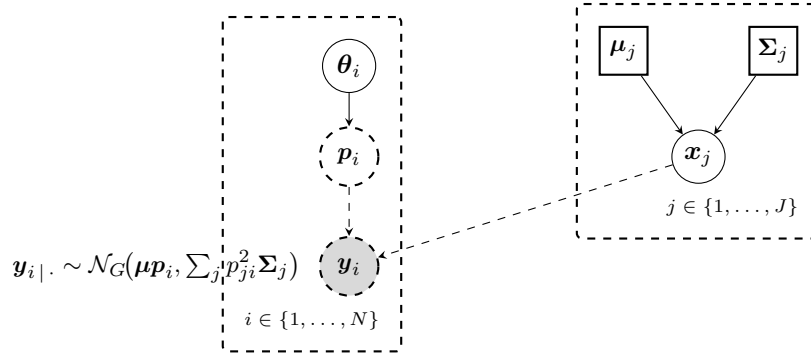
If the matrix is not strictly positive-definite, then some of the diagonal elements can display negative values, otherwise that the individual variances for some of the covariates are negative. It is not physically possible and leads to improper, degenerate probability distributions.



a Standard linear deconvolution (gold-standard template).



b Unconstrained DeCovarT generative model.



c Constrained DeCovarT (regularised soft-max on the simplex).

Fig. 2 Directed graphical models (plate notation), following the conventions of [RevBayes](#). Shaded nodes are observed; square nodes are fixed (plug-in) parameters; dashed circles denote deterministic or constrained quantities. **a** Gold-standard linear model $y_{gi} | . \sim \mathcal{N}(\sum_j x_{gj} p_{ji}, \sigma_i^2)$. **b** Unconstrained DeCovarT convolution $y_i | . \sim \mathcal{N}_G(\mu p_i, \sum_j p_{ji}^2 \Sigma_j)$ with $x_j \sim \mathcal{N}_G(\mu_j, \Sigma_j)$. **c** Constrained DeCovarT: unconstrained coordinates $\theta_i \in \mathbb{R}^{J-1}$ are mapped to the open simplex by the C^2 -diffeomorphism $\psi : \theta \mapsto p$ with $p_j = e^{\theta_j} / (\sum_{k < J} e^{\theta_k} + 1)$ for $j < J$, $p_J = 1 / (\sum_{k < J} e^{\theta_k} + 1)$, and inverse $\psi^{-1}(p) = (\ln(p_j / p_J))_{j < J}$ (Eq. [B11](#)). For identifiability we treat variability as arising from the stochastic covariates x_j .

B.3 Matrix calculus

Fundamental algebra calculus formulas used to derive first-order and second-order derivatives of the generative model of DeCovarT are reported in 3 and 4, respectively.

Remark 3 (First-order matrix calculus) Given two invertible matrices, $A = A(p)$ and $B = B(p)$, functions of a scalar variable p , the following matrix calculus hold:

$$(a) \frac{\partial \text{Det}(\mathbf{A})}{\partial p} = \text{Det}(\mathbf{A}) \text{Tr} \left(\mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial p} \right) \quad (b) \frac{\partial \mathbf{U} \mathbf{A} \mathbf{V}}{\partial p} = \mathbf{U} \frac{\partial \mathbf{A}}{\partial p} \mathbf{V} \quad (c) \frac{\partial \mathbf{A}^{-1}}{\partial p} = -\mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial p} \mathbf{A}^{-1}$$

From a) and fundamental linear algebra properties, we can readily compute applying the chain rule property on the logarithm:

$$\frac{\partial \log(\text{Det}(\mathbf{A}))}{\partial p} = \text{Tr} \left(\mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial p} \right) \\ \frac{\partial \log(\text{Det}(\mathbf{A}^{-1}))}{\partial p} = -\text{Tr} \left(\mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial p} \right)$$

Finally, injecting these first-order matrix derivatives, we obtain:

$$\frac{\partial (\mathbf{y} - \mathbf{x}p)^\top \Theta (\mathbf{y} - \mathbf{x}p)}{\partial p} = -2(\mathbf{y} - \mathbf{x}p)^\top \Theta \mathbf{x} \\ = -2\mathbf{x}^\top \Theta (\mathbf{y} - \mathbf{x}p) \\ \text{with } \mathbf{A} = \mathbf{D} = -\mathbf{x} \in \mathbb{R}^G, \quad \mathbf{b} = \mathbf{e} = \mathbf{y}, \quad \mathbf{C} = \Theta \text{ symmetric}$$

Remark 4 (Second-order matrix calculus) Given an invertible matrix $\mathbf{A}$ depending on a variable p , the following calculus formulas hold:

$$(a) \frac{\partial^2 \mathbf{A}^{-1}}{\partial p_i \partial p_j} = \mathbf{A}^{-1} \left(\frac{\partial \mathbf{A}}{\partial p_i} \mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial p_j} - \frac{\partial^2 \mathbf{A}}{\partial p_i \partial p_j} + \frac{\partial \mathbf{A}}{\partial p_j} \mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial p_i} \right) \mathbf{A}^{-1} \quad (b) \frac{\partial \text{Tr}(\mathbf{A})}{\partial p_i} = \text{Tr} \left(\frac{\partial \mathbf{A}}{\partial p_i} \right)$$

Combining 3 with the linear property of the trace operator yields:

$$\frac{\partial^2 \log(\text{Det}(\mathbf{A}^{-1}))}{\partial^2 p} = -\text{Tr} \left[\mathbf{A}^{-1} \frac{\partial^2 \mathbf{A}}{\partial^2 p_i} \right] + \text{Tr} \left[\left(\mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial p_i} \right)^2 \right]$$

B.4 First and second-order derivation of constrained DeCovarT

B.5 Review of constrained optimisation approaches

Constrained optimization can be addressed either by eliminating constraints through re-parametrisation, softly enforcing them via penalties or barriers, or exactly enforcing feasibility using projections, Lagrangian, or KKT-based methods. Re-parametrisation and penalties are simple but poorly conditioned; barrier and interior-point methods are powerful for convex problems; proximal and projection methods excel in large-scale structured settings; and Lagrangian-based approaches (ALM, SQP, MMA) provide the most general and theoretically grounded framework for high-accuracy solutions.

To reparametrise the log-likelihood function (6) in order to explicitly handling the unit simplex constraint (3), we consider the following mapping function: $\boldsymbol{\psi} : \boldsymbol{\theta} \rightarrow \boldsymbol{p} \mid \boldsymbol{\theta} \in \mathbb{R}^{J-1}, \boldsymbol{p} \in]0, 1[^J$ (B11):

$$1. \quad \quad \quad 2. \quad \boldsymbol{\theta} = \boldsymbol{\psi}^{-1}(\boldsymbol{p}) = \left(\ln \left(\frac{p_j}{p_J} \right) \right)_{j \in \{1, \dots, J-1\}}$$

$$\boldsymbol{p} = \boldsymbol{\psi}(\boldsymbol{\theta}) = \begin{cases} p_j = \frac{e^{\theta_j}}{\sum_{k < J} e^{\theta_k} + 1}, & j < J \\ p_J = \frac{1}{\sum_{k < J} e^{\theta_k} + 1} \end{cases} \quad (\text{B11})$$

that is a C^2 -diffeomorphism, since $\boldsymbol{\psi}$ is a bijection between $\boldsymbol{p}$ and $\boldsymbol{\theta}$ twice differentiable.

Its Jacobian, $\mathbf{J}_{\boldsymbol{\psi}} \in \mathcal{M}_{J \times (J-1)}$ is given by B12:

$$\mathbf{J}_{i,j} = \frac{\partial p_i}{\partial \theta_j} = \begin{cases} \frac{e^{\theta_i} B_i}{A^2}, & i = j, i < J \\ \frac{-e^{\theta_j} e^{\theta_i}}{A^2}, & i \neq j, i < J \\ \frac{-e^{\theta_j}}{A^2}, & i = J \end{cases} \quad (\text{B12})$$

with i indexing vector-valued $\boldsymbol{p}$ and j indexing the first-order order partial derivatives of the mapping function, $A = \sum_{j' < J} e^{\theta_{j'}} + 1$ the sum over exponential (denominator of the mapping function) and $B = A - e^{\theta_i}$ the sum over ratios minus the exponential indexed with the currently considered index i .

The Hessian of the multi-dimensional mapping function $\boldsymbol{\psi}(\boldsymbol{\theta})$ exhibits symmetry for each cell ratio component j , as anticipated in accordance with Schwarz's theorem. It is a third-order tensor of rank $(J-1)(J-1)J$, given by B13:

$$\frac{\partial^2 p_i}{\partial k \partial j} = \begin{cases} \frac{e^{\theta_i} e^{\theta_l} (-B_i + e^{\theta_i})}{A^3}, (i < J) \wedge ((i \neq j) \oplus (i \neq k)) & (a) \\ \frac{2e^{\theta_i} e^{\theta_j} e^{\theta_k}}{A^3}, (i < J) \wedge (i \neq j \neq k) & (b) \\ \frac{e^{\theta_i} e^{\theta_j} (-A + 2e^{\theta_j})}{A^3}, (i < J) \wedge (j = k \neq i) & (c) \\ \frac{B_i e^{\theta_i} (B_i - e^{\theta_i})}{A^3}, (i < J) \wedge (j = k = i) & (d) \\ \frac{e^{\theta_j} (-A + 2e^{\theta_j})}{A^3}, (i = J) \wedge (j = k) & (e) \\ \frac{2e^{\theta_j} e^{\theta_k}}{A^3}, (i = J) \wedge (j \neq k) & (f) \end{cases} \quad (B13)$$

with i indexing $\mathbf{p}$, j and k respectively indexing the first-order and second-order partial derivatives of the mapping function with respect to $\boldsymbol{\theta}$. In line (a), $\oplus$ refers to the Boolean XOR operator, $\wedge$ to the AND operator and $l = \{j, k\} \setminus i$.

To derive the log-likelihood function in 7, we reparametrise $\mathbf{p}$ to $\boldsymbol{\theta}$, using a standard *chain rule formula*. Considering the original log-likelihood function, 6, and the mapping function, B11, the differential at the first order and at the second order is given by B14 and B15, respectively defined in $\mathbb{R}^{J-1}$ and $\mathcal{M}_{(J-1) \times (J-1)}$:

$$\left[\frac{\partial \ell_{\mathbf{y}|\boldsymbol{\zeta}}}{\partial \theta_j} \right]_{j < J} = \sum_{i=1}^J \frac{\partial \ell_{\mathbf{y}|\boldsymbol{\zeta}}}{\partial p_i} \frac{\partial p_i}{\partial \theta_j} \quad (B14)$$

$$\left[\frac{\partial^2 \ell_{\mathbf{y}|\boldsymbol{\zeta}}}{\partial \theta_k \partial \theta_j} \right]_{j < J, k < J} = \sum_{i=1}^J \sum_{l=1}^J \left(\frac{\partial p_i}{\partial \theta_j} \frac{\partial^2 \ell_{\mathbf{y}|\boldsymbol{\zeta}}}{\partial p_i \partial p_l} \frac{\partial p_l}{\partial \theta_k} \right) + \sum_{i=1}^J \left(\frac{\partial \ell_{\mathbf{y}|\boldsymbol{\zeta}}}{\partial p_i} \frac{\partial^2 p_i}{\partial \theta_k \partial \theta_j} \right) \quad (d) \quad (B15)$$

B.6 Numerical optimisation algorithms

The extremum, and by extension the MLE, is a root of the gradient of the log-likelihood. However, in our generative framework, the inverse function cancelling the gradient of 6 is non-closed. Instead, iterated numerical optimisation algorithms that consider first or second-order approximations of the function to optimise are used to approximate the roots.

The *Levenberg-Marquardt (LM)* algorithm ([30]) bridges the gap between between the steepest descent method (first-order) and the Newton-Raphson method (second-order) by inflating the diagonal terms of the Hessian matrix. Far from the endpoint, a second-order descent is favoured for its faster convergence pace, while the steepest approach is privileged close to the extremum since it allows careful refinement of the step size. Specially, we used the LM implementation of R package `marqLevAlg` to infer estimates of the cellular ratios from the bootstrap simulations ([31]). It notably includes an additional convergence criteria, the relative distance to the maximum (RDM), that sets apart extrema from spurious saddle points.

B.7 Heuristic accelerated convergence towards optimal cellular composition

In this section, we *plug-in* the averaged transcriptomic expression per cell type μ as a rough estimate of the hidden sample-specific purified expression profile $\mathbf{X}_{.i}$.

Appendix C Bayesian extension for Modelling the Joint Distribution of Purified Profiles and Ratios

We introduce two variations of the MCMC algorithm, namely the *Metropolis-Hasting* and the *Gibbs sampling* algorithms. They are respectively tailored to approximate distributions for which no explicit form is known (4) or streamline the optimisation of strongly dependent parameters (5).

C.1 An Introduction to Gibbs and Metropolis Hasting Samplers

Definition 4 (Metropolis-Hasting algorithm) First of all, we introduce some key notations:

- Function $f(\theta|\cdot)$ is the *target distribution*, usually the posterior distribution that encompasses both prior knowledge and new data.
- The distribution $q(\theta|\theta^{(q-1)})$ is the *proposal distribution*, alternatively known as the *transition kernel*, and the transient value sampled from it, $\theta^{(*)}$, is the *proposal estimate*.
- The probability of accepting the proposal is naturally called the *acceptance probability*.

Each iteration, indexed by (q) , of the MH algorithm includes the following steps:

1. Draw a proposal, $\theta^{(*)}$, from conditional distribution $q(\theta|\theta^{(q-1)})$.
2. Compute the acceptance probability parameter, noted α :

$$K(\theta^{(*)}) = \min \left(\frac{f(\theta^{(*)}|\cdot)}{f(\theta^{(q-1)}|\cdot)} \frac{q(\theta|\theta^{(q-1)})}{q(\theta^{(q-1)}|\theta)}, 1 \right) \quad (\text{C16})$$

Typically, choices of the acceptance probability and the kernel distribution are tailored to satisfy the balance condition of the MCMC chains and ensure that the chain behaviour reproduces the sampling pattern of the desired distribution.

3. The decision of whether to accept or reject the new state proposal is determined by the previously computed acceptance probability parameter, denoted as α . This parameter sets the threshold for accepting a value drawn from a standard Uniform distribution, $u \sim \mathbb{U}[0, 1]$. This mathematical protocol is further described in C17:

$$\theta^{(q)} = \begin{cases} \theta^{(*)}, & u \leq K(\theta^{(*)}) \\ \theta^{(q-1)}, & u > K(\theta^{(*)}) \end{cases} \quad (\text{C17})$$

In the first case, we say that the proposal is *accepted*, while in the second case, it is *rejected*.

One of the major advantages of the Metropolis-Hastings (MH) algorithm lies in the design of its acceptance probability function, denoted as $K(\theta^{(*)})$. Indeed, by involving the computation of a ratio between two density functions, the normalisation constant, which represents the

value of the marginal likelihood and is usually intractable to compute is naturally cancelled out.

In our modelling framework, the acceptance function further simplifies with the choice of proposing a new proposal by adding an error term following a null-centred, multivariate and symmetric distribution (5):

Remark 5 (Random walk Metropolis-Hastings) Indeed, from the analytical properties of the kernel distribution (C18 and C19):

$$q(\theta^{(*)}|\theta^{(q-1)}) = q(\epsilon) \quad (\text{C18})$$

$$q(\theta^{(q-1)}|\theta^{*}) = q(-\epsilon)$$

$$q(\epsilon) = q(-\epsilon) \quad (\text{C19})$$

, the acceptance probability function simplifies to C20:

$$K(\theta^{(*)}) = \min \left(\frac{f(\theta^{(*)}|\cdot)}{f(\theta^{(q-1)}|\cdot)}, 1 \right) \quad (\text{C20})$$

Definition 5 (Gibbs sampling) The fundamental concept behind Gibbs sampling is to break down the joint posterior distribution of the parameters, into a product of conditional distributions of the parameters. To that end, it is generally assumed that there exists a natural partition of the hidden parameters allowing them to factorise in a meaningful way.

It is usually implemented when the joint conditional posterior distribution is intractable to compute, whereas the conditional distribution for a subset of the parameters, conditioned on all others, is rather straightforward to derive. This is especially the case when the set of hidden parameters is linked to each other, and that numerous numerical constraints linking them must be endorsed.

Using our notations, the following joint posterior distribution $f(\mathbf{p}, \mathbf{X}|\mathcal{D})$, with $\mathcal{D}$ denoting the observed data, here $\mathbf{y}$ is analytically complex to derive, while $f(\mathbf{p}|\mathbf{X}, \mathcal{D})$ and $f(\mathbf{X}|\mathbf{p}, \mathcal{D})$, the posterior cellular ratios and purified individual cell expression profiles, respectively, can be simply computed. In practice, like any MCMC framework, you start to initiate the values for all the parameters. Then, the iterated Gibbs process samples each parameter (or subset of parameters), one at a time, and updates its value conditioned on the other parameters at their current values. It can be proven that after a sufficient number of iterations, the corresponding Markov chain of parameters converges and approximates well the desired joint distribution.

C.2 Pseudo-code Gibbs sampler

We detail below the pseudo-code to generate MCMC chains of the joint distribution of the parameters of interest, in which variable q denotes the running index, B the number of *burn-in* iterations to be discarded after sampling, and Q the actual length of the resulting Markov chain.

Algorithm 2 Pseudo-code of the iterated optimisation method, to retrieve the parameters of DeCovarT’s generative model.

Require: Prior estimates of the mean, $[\boldsymbol{\mu}_{ji} \in \mathbb{R}^G]$, and covariance, $[\boldsymbol{\Sigma}_{ji} \in \mathcal{M}_{\mathbb{R}^G \times G}]$, for each cell population $j \in \{1, \dots, J\}$

Require: Initial estimates of cellular ratios, $\mathbf{p}_{0i}$, and purified cell expression profiles, $\mathbf{X}_{0i}$, for each individual, satisfying both the fundamental linear deconvolution assumption (2) and the unit-simplex constraint (3)

Require: Standard deviations $\sigma_{0(\boldsymbol{\rho})}$ and $\sigma_{0(\mathbf{X})}$ of the additional residual terms added to each cellular ratio and each individual cell profile of the *transition kernel*, respectively

[32], [33], [34] and [35] suggest tuning these hyper-parameters such that the long-run acceptance rates remain between 0.234 and 0.574

```

1: for  $q = 1$  to  $B + Q$  do
2:   for  $i = 1$  to  $N$  do
3:      $\epsilon_{i(\boldsymbol{\rho})} \sim \mathcal{N}_{J-1}(0, \sigma_{0(\boldsymbol{\rho})}^2 \mathbb{I}_{J-1})$ 
4:      $\boldsymbol{\rho}_i^{(*)} \leftarrow \boldsymbol{\rho}_i^{(q-1)} + \epsilon_{i(\boldsymbol{\rho})}$ 
5:     for  $j = 1$  to  $J - 1$  do
6:        $u \sim \mathbb{U}(0, 1)$ 
7:       if  $u < \min\left(1, K_{\boldsymbol{\rho}}\left(\boldsymbol{\rho}_j^{(q-1)} \rightarrow \boldsymbol{\rho}_j^{(*)}\right)\right)$  then
8:          $\rho_{ji}^{(q)} \leftarrow \rho_{ji}^{(*)}$ 
9:       else
10:         $\rho_{ji}^{(q)} \leftarrow \rho_{ji}^{(q-1)}$ 
11:      end if
12:       $\mathbf{p}_i^{(q)} \leftarrow \boldsymbol{\psi}(\boldsymbol{\rho}_i^{(q)})$  (i)
13:      MCMC. $\mathbf{p}_i \leftarrow \mathbf{p}_i^{(q)}$ 
14:    end for
15:    for  $j = 1$  to  $J - 1$  do
16:       $\epsilon_{ji(\mathbf{X})} \sim \mathcal{N}_G(0, \sigma_{0(\mathbf{X})}^2 \mathbb{I}_G)$ 
17:       $\mathbf{x}_{ji}^{(*)} \leftarrow \mathbf{x}_{ji}^{(q-1)} + \epsilon_{ji(\mathbf{X})}$ 
18:       $u \sim \mathbb{U}(0, 1)$ 
19:      if  $u < \min\left(1, K_{\mathbf{x}}\left(\mathbf{x}_{ji}^{(q-1)} \rightarrow \mathbf{x}_{ji}^{(*)}\right)\right)$  then
20:         $\mathbf{x}_{ji}^{(q)} \leftarrow \mathbf{x}_{ji}^{(*)}$ 
21:      else
22:         $\mathbf{x}_{ji}^{(q)} \leftarrow \mathbf{x}_{ji}^{(q-1)}$ 
23:      end if
24:    end for
25:     $\mathbf{x}_{Ji}^{(q)} \leftarrow \frac{\mathbf{y}_i - \sum_{j=1}^{J-1} \mathbf{x}_{ji}^{(q)} p_{ji}^{(q)}}{p_{Ji}^{(q)}}$  (ii)
26:    MCMC. $\mathbf{X}_i \leftarrow \mathbf{X}_i^{(q)}$ 
27:  end for
28: end for

```

We implemented two reparametrisations at each global iteration (outer loop) to ensure that the kernel distribution generates proposals that fall within the “support” of the target distribution, and notably the adhesion to the fundamental linear deconvolution relation (2) and the unit-simplex constraint over the cellular ratios (3):

1. First, the mapping function, described in (i) and in B11, enforces that the estimated cellular ratios adhere to the unit-simplex constraint (3).
2. Second, the fundamental linearity of deconvolution, 2, is endorsed by the update formula (ii). Formula 2 implies that the last J cellular expression profile, $\mathbf{x}_{.j}$ is not a free parameter, and can be rewritten as a combination of the others, given by C21:

$$\begin{aligned} \mathbf{y}_i &= \sum_{j=1}^{J-1} p_{ji}^{(q)} \mathbf{x}_{ji}^{(q)} + p_{Ji}^{(q)} \mathbf{x}_{Ji}^{(q)} = \sum_{j=1}^J p_{ji}^{(q-1)} \mathbf{x}_{ji}^{(q-1)} \\ &\iff \\ \mathbf{x}_{Ji}^{(q)} &= \frac{\mathbf{y}_i - \sum_{j=1}^{J-1} \mathbf{x}_{ji}^{(q)} p_{ji}^{(q)}}{p_{Ji}^{(q)}} \end{aligned} \tag{C21}$$

C.3 Derivation of the Acceptance Probability Function

By utilising a Random Walk MH approach, additionally cancelling out the normalisation constant (C20), the acceptance probability function to compute is simply the product of the prior distributions and the likelihood of the observed data, $f(\boldsymbol{\theta}) \times f(\mathcal{D}|\boldsymbol{\theta})$.

To simplify further this product of distributions, we preliminary suppose that the density distribution characterising the priors is improper, in other words, that $f(\boldsymbol{\theta})$ is always equal to 1 on $\boldsymbol{\Theta}$. As we provide a closed form of the log-likelihood of our generative model (6), and not the likelihood, we need to apply an exponential transformation to recover the desired acceptance probability (C22):

$$K(\theta^{(*)}) = \min \left(1, \exp \left(\ell(\theta^{(*)}|\cdot) - \ell(\theta^{(q-1)}|\cdot) \right) \right) \tag{C22}$$

with:

- $\boldsymbol{\theta}^{(*)}$, the current proposal (either $\mathbf{p}^{(*)}$ or $\mathbf{X}^{(*)}$)
- $\ell(\boldsymbol{\theta}^{(*)}|\cdot) = \log \left(f(\mathbf{Y}|\boldsymbol{\theta}^{(*)}, \boldsymbol{\zeta}) \right)$, the log-likelihood of the currently observed data, conditioned on the current values of the latent parameters to estimate and the user-defined parameters $\boldsymbol{\zeta}$, given in our generative model by 6.
- $\boldsymbol{\zeta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\mu} = (\boldsymbol{\mu}_{.j})_{j \in \tilde{J}} \in \mathcal{M}_{G \times J}$, $\boldsymbol{\Sigma} = \sum_{j=1}^J p_j^2 \boldsymbol{\Sigma}_j \in \mathcal{M}_{G \times G}$ denote the parameters provided by the user before conducting the study.

Appendix D Miscellaneous studies

D.1 Simplifications adopting the marker-based paradigm

A simpler case to derive the log-likelihood is adopting a marker-based paradigm, as we can factorise the multivariate conditional distribution as the product of J conditional independent multivariate Gaussian distributions, each defined on a known gene marker set G_j and with only parameter to estimate p_j , formally D23:

$$P_{\mathbf{p}}(\mathbf{Y}|\mathbf{X}, \Sigma) = \prod_{j=1}^J P_{p_j}(y_{G_j}|\mathbf{x}_{G_j}, \Sigma_j) \quad (\text{D23})$$

The log-likelihood in a marker-based context is given by D24:

$$\begin{aligned} \ell(\mathbf{p}|\mathbf{y}, \mathbf{X}, \Sigma) &= \sum_{j=1}^J \mathbb{P}_{p_j}(y_{G_j}|\mathbf{x}_{G_j}, \Sigma_j) \\ &= C + \sum_{j=1}^J \left[-|G_j| \log(p_j) - \frac{1}{2p_j^2} (y_{G_j} - \mathbf{x}_{G_j} p_j) \Theta_j (y_{G_j} - \mathbf{x}_{G_j} p_j)^\top \right] \quad (\text{a}) \\ &= C + \sum_{j=1}^J \left[-|G_j| \log(p_j) - \frac{1}{2p_j^2} \sum_{i=g}^{G_j} \Theta_{gg} \epsilon_{gg}^2 - \frac{1}{p_j^2} \sum_{1 \leq g < l \leq |G_j|} \Theta_{gl} \epsilon_{gl}^2 \right] \quad (\text{b}) \end{aligned} \quad (\text{D24})$$

with $C = \frac{1}{2} \sum_{j=1}^J \log(\det(\Theta_j)) + \frac{1}{2} GJ \log(2\pi)$ a constant, $\epsilon_{gg} = (y_g - p_j x_g)$ the residual of canonical linear regression of gene g in market set G_j , $\epsilon_{gl} = (y_g - p_j x_g)(y_l - p_j x_l)$, $g \neq l$ and Θ_{gl} input of the precision matrix indexed by row index g and column index l .

Now, computing the first-order derivative of the log-likelihood using matrix calculus 4 and setting to 0, we get D25:

$$\begin{aligned} \frac{\partial \ell(\mathbf{p}|\mathbf{y}, \mathbf{X}, \Sigma)}{\partial p_j} &= 0 \\ &\iff \\ -\frac{|G_j|}{p_j} + \frac{(y_{G_j} - \mathbf{x}_{G_j} p_j)^\top \Theta_j (y_{G_j} - \mathbf{x}_{G_j} p_j)}{p_j^3} + \frac{\mathbf{x}_{G_j}^\top \Theta_j (y_{G_j} - \mathbf{x}_{G_j} p_j)}{p_j^2} &= 0 \\ &\iff \quad (\text{D25}) \\ -|G_j| + \frac{y_{G_j}^\top \Theta_j y_{G_j}}{p_j^2} - 2 \frac{\mathbf{x}_{G_j}^\top \Theta_j y_{G_j}}{p_j} + \cancel{\mathbf{x}_{G_j}^\top \Theta_j \mathbf{x}_{G_j}} + \frac{\mathbf{x}_{G_j}^\top \Theta_j y_{G_j}}{p_j} - \cancel{\mathbf{x}_{G_j}^\top \Theta_j \mathbf{x}_{G_j}} &= 0 \\ &\iff \\ -|G_j| p_j^2 - \mathbf{x}_{G_j}^\top \Theta_j y_{G_j} p_j + y_{G_j}^\top \Theta_j y_{G_j} &= 0 \end{aligned}$$

Solving quadratic equation D25 yields two real roots corresponding to two extrema of the log-likelihood function:

$$(p_{j1}, p_{j2}) = \frac{-\mathbf{x}_{G_j}^\top \Theta_j y_{G_j} \pm \sqrt{\left(\mathbf{x}_{G_j}^\top \Theta_j y_{G_j}\right)^2 + 4|G_j| y_{G_j}^\top \Theta_j y_{G_j}}}{2|G_j|} \quad (\text{D26})$$

Indeed, $\Delta = \left(\mathbf{x}_{G_j}^\top \Theta_j y_{G_j}\right)^2 + 4|G_j| y_{G_j}^\top \Theta_j y_{G_j}$ is strictly positive, with first term of the discriminant squared raised and second term strictly positive from the condition of positiveness imposed on the covariance matrix 3.

To identify the MLE, we should check which of the two solutions is a maximum, by computing the second-order derivative of the log likelihood with respect to $\hat{p}$ and assert which one is negative. The second derivative is given by the following formula D27:

$$\frac{\partial^2 \ell(\mathbf{p}|\mathbf{y}, \mathbf{X}, \Sigma)}{\partial^2 p_j} = \frac{|G_j|}{p_j^2} - 3 \frac{y_{G_j}^\top \Theta_j y_{G_j}}{p_j^4} + 2 \frac{x_{G_j}^\top \Theta_j y_{G_j}}{p_j^3} \quad (\text{D27})$$

associated to the two following real roots D28:

$$\frac{-x_{G_j}^\top \Theta_j y_{G_j} \pm \sqrt{\left(x_{G_j}^\top \Theta_j y_{G_j}\right)^2 + 3|G_j| y_{G_j}^\top \Theta_j y_{G_j}}}{|G_j|} \quad (\text{D28})$$

Using the property that a quadratic function (we make it appeared by multiplying by the always positive factor p^4) has an opposite sign between its roots to that of its quadratic term (in D27, the always positive term $|G_j|$), we deduce that the MLE estimate, $\hat{p}_j$, as computed in D26 is indeed a maximum if it's included in the interval defined in D28.

D.2 Robust linear regression with GLS

An ad-hoc approximation could be made, by assuming fixed and equi-balanced cellular proportions: $\sum_{j=1}^J p_j = \frac{1}{J}$ to estimate the global covariance of the conditional distribution. Using this hypothesis, the global covariance is given by D29:

$$\text{Cov}(\mathbf{y}|\mathbf{X}) = \frac{1}{J^2} \sum_{j=1}^J \Sigma_j \quad (\text{D29})$$

Indirectly, we consider making this hypothesis that the the variance of purified expression profiles is fixed, and independent from the magnitude of their proportions in the sample. These are exactly the hypotheses made by the *generalised least squares method*, whose properties of the corresponding estimate are described in 6.

Theorem 6 (Generalised Least Squares Regression) *Modelling the multivariate linear conditional problem, stating that:*

1. $\mathbf{y} = \mathbf{X}\mathbf{p} + \epsilon$
2. $\mathbb{E}(\epsilon|\mathbf{X}) = 0$
3. $\text{Cov}(\epsilon|\mathbf{X}) = \Sigma$

with σ a known non-singular covariance matrix. The corresponding GLS estimate that minimises the squared Mahalanobis distance of the residuals is given by:

$$\hat{\mathbf{p}}^{GLS} = (\mathbf{X}^\top \Theta \mathbf{X})^{-1} \mathbf{X}^\top \Theta \mathbf{y}$$

where Θ is the precision matrix, inverse of the covariance matrix. $\hat{\mathbf{p}}^{GLS}$ is an unbiased, consistent, efficient and asymptotically normal estimator. Assuming additionally that the conditional distribution is modelled by a multivariate Gaussian distribution, with mean $\mathbf{X}\mathbf{p}$ and covariance Σ , the associated MLE estimate is the same as the GLS.

Of note, by using a Cholesky decomposition $\Sigma = \mathbf{C}\mathbf{C}^\top$ and multiply the observations and the predicted values by $\mathbf{C}^{-1}$, we can turn the problem in a standard OLS optimisation, removing the heteroscedascity and the covariance between the observations. Using then 5 to this transformed data, we show that the GLS estimate is the best linear unbiased estimator of $\mathbf{p}$.⁴

References

- [1] Newman, Aaron and Liu, Chih and others. Robust Enumeration of Cell Subsets from Tissue Expression Profiles. *Nature methods* (2015).
- [2] Gong, Ting and Szustakowski, Joseph D. DeconRNASeq: A Statistical Framework for Deconvolution of Heterogeneous Tissue Samples Based on mRNA-Seq Data. *Bioinformatics (Oxford, England)* (2013).
- [3] Chiquet, J., Mariadassou, M. & Robin, S. Variational inference for sparse network reconstruction from count data (2018). URL <http://arxiv.org/abs/1806.03120>.
- [4] Batardiere, B., Chiquet, J. & Mariadassou, M. Evaluating Parameter Uncertainty in the Poisson Lognormal Model with Corrected Variational Estimators (2024). URL <http://arxiv.org/abs/2411.08524>.
- [5] Ogier du Terrail, J. *et al.* FedECA: Federated external control arms for causal inference with time-to-event data in distributed settings. *Nature Communications* (2025). URL <https://www.nature.com/articles/s41467-025-62525-z>.
- [6] Melnykov, V., Chen, W.-C. & Maitra, R. MixSim: An R Package for Simulating Data to Study Performance of Clustering Algorithms. *Journal of Statistical Software* (2012). URL <https://doi.org/10.18637/jss.v051.i12>.
- [7] Qiu, W. & Joe, H. Separation index and partial membership for clustering. *Computational Statistics & Data Analysis* (2006). URL <https://www.sciencedirect.com/science/article/pii/S0167947304002749>.
- [8] Wang, X., Park, J., Susztak, K., Zhang, N. R. & Li, M. Bulk tissue cell type deconvolution with multi-subject single-cell expression reference. *Nature Communications* **10**, 380 (2019). URL <https://www.nature.com/articles/s41467-018-08023-x>.
- [9] Mazumder, Rahul and Hastie, Trevor. The Graphical Lasso: New Insights and Alternatives. *Electronic Journal of Statistics* (2011).

⁴Weighted least squares (WLS) estimation is a sub-problem of generalised least squares optimisation, in which the off-diagonal entries of the covariance matrix are set to zero, and the heteroscedascity of each observation is encoded in the diagonal terms.

- [10] Fornberg, B. Numerical differentiation of analytic functions. *ACM Trans. Math. Softw.* (1981). URL <https://doi.org/10.1145/355972.355979>.
- [11] Zaitsev, Aleksandr and Chelushkin, Maksim and others. Precise Reconstruction of the TME Using Bulk RNA-seq and a Machine Learning Algorithm Trained on Artificial Transcriptomes. *Cancer Cell* (2022).
- [12] Racle, Julien and de Jonge, Kaat and others. Simultaneous Enumeration of Cancer and Immune Cell Types from Bulk Tumor Gene Expression Data. *eLife* (2017).
- [13] Newman, A. M. *et al.* Determining cell type abundance and expression from bulk tissues with digital cytometry. *Nature Biotechnology* (2019).
- [14] Finotello, Francesca and Mayer, Clemens and others. Molecular and Pharmacological Modulators of the Tumor Immune Contexture Revealed by Deconvolution of RNA-seq Data. *Genome Medicine* (2019).
- [15] Monaco, G. *et al.* RNA-Seq Signatures Normalized by mRNA Abundance Allow Absolute Deconvolution of Human Immune Cell Types. *Cell Reports* (2019).
- [16] Ritchie, M. E. *et al.* Limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Research* (2015). URL <https://doi.org/10.1093/nar/gkv007>.
- [17] Robinson, M. D., McCarthy, D. J. & Smyth, G. K. edgeR: A Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics* (2010). URL <https://doi.org/10.1093/bioinformatics/btp616>.
- [18] Varet, H., Brillet-Gueguen, L., Coppee, J.-Y. & Dillies, M.-A. SARTools: A DESeq2- and EdgeR-Based R Pipeline for Comprehensive Differential Analysis of RNA-Seq Data. *PLOS ONE* (2016). URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0157022>.
- [19] Chion, M. & Leroy, A. A Bayesian Framework for Multivariate Differential Analysis accounting for Missing Data. *The Annals of Applied Statistics* (2023). URL <http://arxiv.org/abs/2307.08975>.
- [20] Chion, M., Carapito, C. & Bertrand, F. Accounting for multiple imputation-induced variability for differential analysis in mass spectrometry-based label-free quantitative proteomics. *PLOS Computational Biology* (2022). URL <http://arxiv.org/abs/2108.07086>.
- [21] Zuo, Yiming and Cui, Yi and others. INDEED: Integrated Differential Expression and Differential Network Analysis of Omic Data for Biomarker Discovery. *Methods (San Diego, Calif.)* (2016).

- [22] Aliee, Hananeh and Theis, Fabian J. AutoGeneS: Automatic Gene Selection Using Multi-Objective Optimization for RNA-seq Deconvolution. *Cell Systems* (2021).
- [23] Barabasi, A.-L. & Albert, R. Emergence of Scaling in Random Networks. *Science* (1999). URL <https://www.science.org/doi/abs/10.1126/science.286.5439.509>.
- [24] Newman, M., Barabasi, A.-L. & Watts, D. J. *The Structure and Dynamics of Networks* (Princeton University Press, 2011).
- [25] Lima, M. The powerful law of the power law and other myths in network biology1. *Molecular BioSystems(MBS)* (2009). URL <https://doi.org/10.1039/B908681A>.
- [26] Holland, P. W., Laskey, K. B. & Leinhardt, S. Stochastic blockmodels: First steps. *Social Networks* (1983). URL <https://www.sciencedirect.com/science/article/pii/0378873383900217>.
- [27] Shafarevich, I. R. & Remizov, A. O. in *Linear Equations* (eds Shafarevich, I. R. & Remizov, A. O.) *Linear Algebra and Geometry* (Springer, Berlin, Heidelberg, 2013). URL https://doi.org/10.1007/978-3-642-30994-6_1.
- [28] Abbas, Alexander R. and Wolslegel, Kristen and others. Deconvolution of Blood Microarray Data Identifies Cellular Activation Patterns in Systemic Lupus Erythematosus. *PloS One* (2009).
- [29] Efron, B. Are a set of microarrays independent of each other? *The Annals of Applied Statistics* (2009). URL <https://projecteuclid.org/journals/annals-of-applied-statistics/volume-3/issue-3/Are-a-set-of-microarrays-independent-of-each-other/10.1214/09-AOAS236.full>.
- [30] Levenberg, Kenneth. A method for the solution of certain non-linear problems in least squares. *Quarterly of Applied Mathematics* (1944). URL <https://www.ams.org/qam/1944-02-02/S0033-569X-1944-10666-0/>.
- [31] Philipps, V., Hejblum, B. P., Prague, M., Commenges, D. & Proust-Lima, C. Robust and Efficient Optimization Using a Marquardt-Levenberg Algorithm with R Package marqLevAlg. *The R Journal* (2021). URL <http://arxiv.org/abs/2009.03840>.
- [32] Siems, T. & Koepfel, L. A note on the Metropolis-Hastings acceptance probabilities for mixture spaces (2019). URL <http://arxiv.org/abs/1808.00789>.
- [33] Andrieu, C., Doucet, A., Yldrm, S. & Chopin, N. On the utility of Metropolis-Hastings with asymmetric acceptance ratio (2018). URL <http://arxiv.org/abs/1803.09527>.

- [34] Vogrinc, J. & Kendall, W. S. Counterexamples for optimal scaling of Metropolis–Hastings chains with rough target densities. *The Annals of Applied Probability* (2021). URL <https://projecteuclid.org/journals/annals-of-applied-probability/volume-31/issue-2/Counterexamples-for-optimal-scaling-of-MetropolisHastings-chains-with-rough-target/10.1214/20-AAP1612.full>.
- [35] Marnissi, Y., Chouzenoux, E., Benazza-Benyahia, A. & Pesquet, J.-C. Majorize-Minimize Adapted Metropolis-Hastings Algorithm. *IEEE Transactions on Signal Processing* (2020). URL <https://hal.science/hal-01909153>.